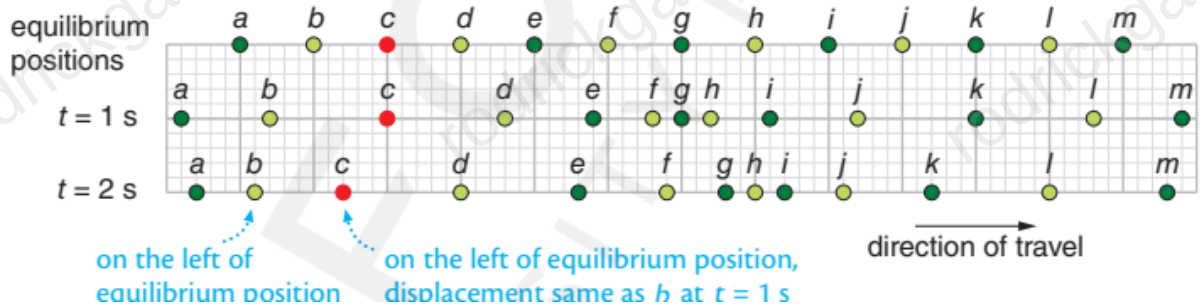
3B07 Sound

3.7.1 Instantaneous particle speed and kinetic energy

For simple harmonic particle motion in a wave, speed is greatest at equilibrium and momentarily zero at maximum displacement before reversal. Since $K = \frac{1}{2}mu^2$, kinetic energy is greatest in the first case and zero in the second. Zero displacement therefore does not mean rest. Particle speed varies during oscillation and differs from the waveform's propagation speed.

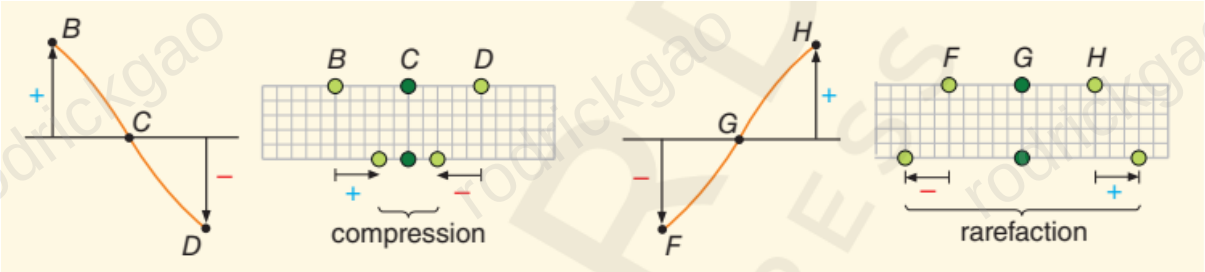


Compare particle motion at equilibrium and at maximum displacement.

Textbook sources: 3B07 · PDF 6

3.7.2 Compression and rarefaction on displacement graphs

First establish that displacement to the right is positive. On a displacement-distance graph, a downward crossing of zero marks a compression centre; an upward crossing marks a rarefaction centre. Graph height represents displacement, not vertical particle motion. The neighbouring-particle diagram shows particles crowding towards a compression or separating away from a rarefaction.



Neighbouring particles crowd together in a compression and move apart in a rarefaction.

Textbook sources: 3B07 · PDF 10, 11

3.7.3 Sound speed and the medium

Sound requires a medium; its speed depends on properties including elasticity, density and temperature. Common materials often transmit sound faster in solids than liquids and faster in liquids than gases, but this is not universal. With medium conditions fixed, increasing loudness or changing source frequency does not normally change sound speed directly.

$$v_{\text{solid}} > v_{\text{liquid}} > v_{\text{gas}}$$

Typical sound-speed ordering for common materials, not a universal inequality for every solid, liquid and gas. v: sound speed (m s^{-1}).

Textbook sources: 3B07 · PDF 14

3.7.4 Reflection, refraction and interference of sound

Sound reflects, refracts, diffracts and interferes like other waves. Echoes result from reflection; regions of different sound speed can bend propagation; spreading around obstacles involves diffraction. Coherent sources can produce alternating loud and quiet positions. Match each observation to its mechanism rather than describing every effect as an echo.

Textbook sources: 3B07 · PDF 22

3.7.5 · 3.7.7 Audible sound, infrasound and ultrasound

A young person with normal hearing typically hears about 20 Hz to 20 kHz. Lower frequencies are infrasound and higher frequencies are ultrasound; actual limits vary with age and individual hearing. This classification concerns frequency, not loudness. Ultrasound is a mechanical wave, not electromagnetic radiation, and cannot travel through a vacuum.

$$20\ \text{Hz} \lesssim f \lesssim 20\,000\ \text{Hz}$$

$$f > 20\,000\ \text{Hz}$$

Approximate audible range for a young healthy human; sensitivity and upper limit vary with age and the individual.

Ultrasound has frequency above approximately 20 kHz, the conventional upper limit of human hearing.

Textbook sources: 3B07 · PDF 25, 26

3.7.6 Applications of ultrasound

Transmitted ultrasound and returning echoes locate fish, scan structures and reveal cracks in materials. Other textbook applications include cleaning delicate objects and concentrating energy to break kidney stones. Explain whether the application uses echo detection or energy delivery rather than saying only that ultrasound has high frequency.

Textbook sources: 3B07 · PDF 26

3.7.8 Pitch, loudness and quality

Pitch generally rises with frequency. At comparable frequency and conditions, greater amplitude usually sounds louder. Quality or timbre depends on waveform and harmonic content, allowing instruments with the same pitch and loudness to sound different. These properties have different physical bases: a high-pitched sound is not necessarily loud.

Textbook sources: 3B07 · PDF 30

3.7.9 Typical sound intensity levels

Textbook examples include 20 dB in a country park, 40 dB for soft conversation, 60 dB for ordinary conversation and 70 dB for a vacuum cleaner 3 m away. Zero dB is a hearing-threshold reference, not the absence of sound. Use any specified distances and measurement conditions when comparing examples.

Textbook sources: 3B07 · PDF 32